

高等数学核心知识速查

第 1 页 · 极限与导数 (含例题)

一、函数与极限

重要极限: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x = e$

等价无穷小 ($x \rightarrow 0$): $\sin x \sim x$, $\tan x \sim x$, $\ln(1+x) \sim x$, $e^x - 1 \sim x$, $1 - \cos x \sim \frac{x^2}{2}$, $(1+x)^\alpha - 1 \sim \alpha x$

二、导数与微分

求导法则: $(uv)' = u'v + uv'$, $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$, 链式 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$$(x^\mu)' = \mu x^{\mu-1}$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x$$

$$(e^x)' = e^x$$

$$(a^x)' = a^x \cdot \ln a$$

$$(\ln x)' = \frac{1}{x}$$

$$(\log_a x)' = \frac{1}{x \ln a}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

$$(\sec x)' = \sec x \cdot \tan x$$

$$(\cot x)' = -\csc^2 x$$

$$(\csc x)' = -\csc x \cdot \cot x$$

$$(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\operatorname{arccot} x)' = \frac{-1}{1+x^2}$$

例题 1 求极限

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \cdot \sin x}$$

解: $x \rightarrow 0$ 时 $1 - \cos x \sim \frac{x^2}{2}$, $\sin x \sim x$, 故原式 $= \lim_{x \rightarrow 0} \frac{x^2/2}{x \cdot x} = \frac{1}{2}$

三、不定积分与定积分

$$\int x^{\mu} dx = \frac{x^{\mu+1}}{\mu+1} + C \quad (\mu \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \frac{dx}{1+x^2} = \arctan x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$\int \tan x dx = -\ln|\cos x| + C$$

$$\text{分部积分: } \int u dv = uv - \int v du$$

牛顿-莱布尼茨:

$$\int_a^b f(x) dx = F(b) - F(a)$$

奇偶性: $\int_{-a}^a f(x) dx = 0$ (f 奇); $= 2 \int_0^a f(x) dx$ (f 偶)

四、无穷级数

比值判别法: $\rho = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$, $\rho < 1$ 收敛, $\rho > 1$ 发散

幂级数收敛半径: $R = \lim_{n \rightarrow \infty} \frac{|a_n|}{|a_{n+1}|}$ ($|x| < R$ 内绝对收敛)

常用展开: $e^x = \sum \frac{x^n}{n!}$, $\frac{1}{1-x} = \sum x^n$ ($|x| < 1$)

五、常微分方程

一阶线性: $y' + P(x)y = Q(x)$, 通解 $y = e^{-\int P(x)dx} [\int Q(x) \cdot e^{\int P(x)dx} dx + C]$

二阶常系数齐次: $y'' + py' + qy = 0$, 特征方程 $r^2 + pr + q = 0$; 两不等实根 $r_1 \neq r_2 \Rightarrow y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$

例题 2 求不定积分 $\int x \cdot e^x dx$

解: 分部积分, 取 $u = x$, $dv = e^x dx$: $\int x \cdot e^x dx = x \cdot e^x - \int e^x dx = (x - 1) \cdot e^x + C$